1.2 Comparison of Unlike Fractions

Think about the situation 1

Murugan has scored $\frac{7}{10}$ in Science and $\frac{9}{10}$ in Mathematics test. In which subject he has performed better? It is quite easy to say his performance is better in Mathematics. But can you find, the better performance of Murugan between the two test scores such as $\frac{9}{10}$ and $\frac{13}{20}$ in Mathematics. We need to convert both the marks as like fractions.

The equivalent fraction of $\frac{9}{10}$ is $\frac{18}{20}$. Now we can compare the first test score with that of the second test score because both the scores are out of 20 marks. Here 18 > 13. So, $\frac{18}{20} > \frac{13}{20}$. Thus, Murugan has performed better in the first test.

Think about the situation 2

In a Hockey tournament, Team A played 6 matches and won 5 matches out of it. Team B played 5 matches and won 4 matches out of it. If both the teams performed consistently in this way, find out which team will win the tournament?

From these we need to see which is greater $\frac{5}{6}$ or $\frac{4}{5}$? How can we find this? The total number of matches played by each team differs. By finding the equivalent fractions of $\frac{5}{6}$ and $\frac{4}{5}$, we can equalize the number of matches played by team A and team B.

$$\frac{5}{6} = \frac{10}{12} = \frac{15}{18} = \frac{20}{24} = \frac{25}{30} \qquad \frac{4}{5} = \frac{8}{10} = \frac{12}{15} = \frac{20}{25} = \frac{24}{30}$$

Note that the common denominator of equivalent fraction is 30, which is 5×6. It is the common multiple of both 5 and 6.

Here $\frac{25}{30} > \frac{24}{30}$. So Team A will win the game.

Note

To compare two or more unlike fractions, we have to convert them into 'like fractions'. These 'like fractions' are the equivalent fractions of the given fractions. The denominator of the 'like fractions' is the Least Common Multiple (LCM) of the denominators of the given unlike fractions.

Example 2 Madhu ate $\frac{2}{5}$ of the chocolate bar and Nandhini ate $\frac{1}{3}$ of the chocolate bar. Who has eaten more?

Solution

The portion of the chocolate eaten by Madhu = $\frac{2}{5}$

The portion of the chocolate eaten by Nandhini = $\frac{1}{3}$

Here the portions of the chocolates eaten by both differ.

6th_ Maths _Chapter 1.indd 5 22-10-2019 18:10:53